Biometric ATM Authentication using Machine Learning

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***Abstract* — ATMs are used by many people these days. But it is difficult to take their ATM card everywhere, people may forget to have their ATM card. The ATM card can get damaged and users can end up in a situation where they cannot access their funds. There aren't any right ones Authentication methods used for security during ATM transactions. In this paper, security approaches from ATM focused and has been enhanced with biometric base authentication technique. The aim is to model an efficient automatic identification system that simplifies the management of monetary transactions using facial recognition and user verification with otp.**

Keywords—ATM, Biometric Authentication,OTP.

I. INTRODUCTION

Due to the rapid development of science and technology, coming creations will be built with strong security. On the other hand On the other hand, threats are also made to break this level of security. However, the improvement in automation has had a positive impact Overall; however, various financial institutions like banks and applications like ATMs are still subject to theft and trickery. Due to the Advances in Technology Fraudsters are finding new ways to attack, increasing security. Biometric technology is a method that is easier to implement and with which a higher level of security can be achieved in many areas. Offer biometric identifiers several advantages over traditional and current types of authentication and security. The newer ATM security authentication technique is entirely dependent on PIN-based verification. Factors such as Emergency, reminds of pins, speed of interaction, accidental pin sharing affects the system in many ways. Cards with magnet Chips are easy to clone. The security and the unguarded are opposite sides of the same coin, an automated machine becomes vulnerable due to the weakness of its security. ATM manufacturers reinforce and add security features ATM, so that the customer can carry out banking transactions easily and without fear of skimming amount from their account and the same scammers are working at a similar rate to crack the innovative security feature so they can access through the ATM to exploit bank customer's accounts. The main motivation of this project is to strengthen the security of the traditional ATM module. We have a new presumption concept that improves the overall experience, usability and convenience of ATM transactions. Features like face recognition and One-Time Password (OTP) are used to strengthen account security and user privacy. Facial recognition technology Help the machine to uniquely recognize each and every user, making the face the master key. So the chances are completely destroyed artifice through theft and duplication of ATM cards. Development of the ATM system using the LRR algorithm for the face Recognition of the user while authenticating the real user of the account. The system will also have the other option for login i.e. another user.

The method for biometric ATM authentication that uses:

- Image Preprocessing: Use of open cv will capture images of user .

II. LITERATURE REVIEW

Praveena.P , Et al [1] propose the Principal component analysis (PCA), which is the heart of the Eigen faces method, finds a linear combination of features that maximizes the total variance in the data. Although this is clearly a powerful way to represent data, it does not consider any class and therefore a lot of discriminating information can be lost when discard components.

Pooja Surawse, Et al [2] as proposed the Area of work is basically intent on Design and Implementation of Face Detection based ATM Security System using LRR algorithm. The first standalone user, where the system will ask for "Detect face" and allow the transaction to continue if it is a matching with the image store in the database of banks, outside of this system, will refuse the transaction after some warnings for Guest. The user, where in the system will ask for "OTP" and allow to process the transaction if the guest user enters the correct OTP that was sent to the authorized user.

Darwin Nesakumar A1, Et al[3] developed a s can be done using a USB camera and a fingerprint module. If analysis matched, the ATM will allow the transaction. Incase if unauthorized person inserts ATM card, fingerprint and image will be checked as they do not match, so their image with an OTP will be sent to account holders mail, only after entering this OTP in this system, it will allow user to withdraw money, otherwise it will stop the process. All control is provided using the Raspberry pi controller. Now let's talk about the processing performed in this security system.

Mala K [4] developed a the integration of biometrics into the banking application has made the ATM transaction system more reliable and secure. The OTP concept added to the system further improves security and saves us from having to remember passwords. Moreover the system is built on embedded technology that makes it user-friendly and non-invasive. Thanks to this system, the ATM terminal is safe from attack by thieves.

Pratiksha Shetiya, Et al [5]. Develops to identifying and comparing a user's face(IRIS) of eyes. Pictures were captured by the database and given to the computer system where it is processed by various MATLAB functions. Then the database iris image was compared to the output iris image and if it matches, the user has access to the account or something else It rejects the user's request. Finally the system that has 94.6% detection rate .

III. METHODOLOGY

The purpose of this paper is to find a Machine Learning model to identify the real time face recognition. In this system, we use the pycharm that makes it easy to set up and access and user interface design to use of flask framework in python.

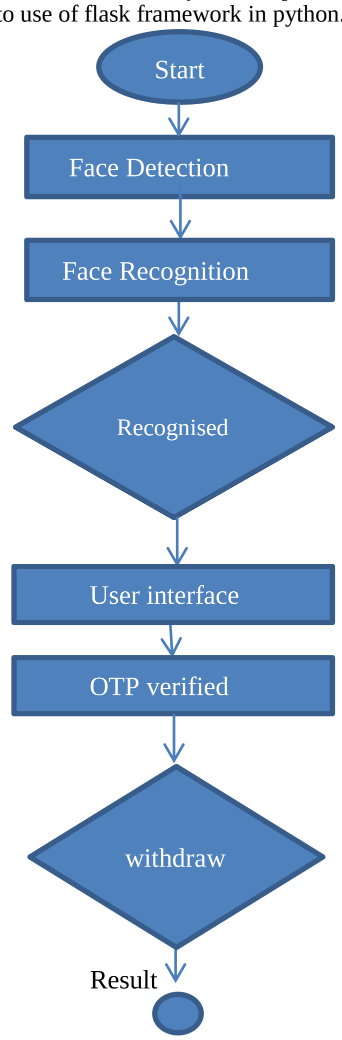

Figure 1: Biometric ATM Authentication   The steps that require to be followed are:

A. Requirements

- Import packages.
- Pycharm
- Images
- Webcam

B. **Import Opencv&Dlib:** I've used OpenCV and dlib a lot for face detection and recognition and dlib is much more accurate than the OpenCV based face detector Haar.

C. **OTP verification:** This is the step that is to identifies the user and we set a email id and password and system will verified the user to the security purpose.

D. **Machine Learning**: Here machine learning is used to recognize the face. The face is identified by frames at real-time from the data array that had trained before.

IV. IMPLEMENTATION

The model building is the main step in the face detection and OTP verification. First step is open a Pycharm flask project then, Import the packages that are necessary.

```python
from flask import Flask, render_template, Response
import cv2
import face_recognition
import numpy as np

app = Flask(__name__)
camera = cv2.VideoCapture(0)
# Load a sample picture and learn how to recognize it.
image = face_recognition.load_image_file("image/jeena (2).jpeg")
face_encoding = face_recognition.face_encodings(image)[0]
```

Figure 2: Import Packages.

```python
camera = cv2.VideoCapture(0)
# Load a sample picture and learn how to recognize it.
image = face_recognition.load_image_file("image/jeena (2).jpeg")
face_encoding = face_recognition.face_encodings(image)[0]
```

Figure 3: Load sample picture and learn and recoganize it.

```python
# Create arrays of known face encodings and their names
known_face_encodings = [
    face_encoding
]
known_face_names = [
    "jeena"
]
# Initialize some variables
face_locations = []
face_encodings = []
face_names = []
process_this_frame = True
```

Figure 4: Creating Array for Known face encoding their names.

```html
app.py   index.html   result.html
<!DOCTYPE html>
<html>


  <body>
    <div class="container">
        <div class="row">
            <div class="col-lg-8  offset-lg-2">
                <h3 class="mt-5">Live Streaming</h3>
                <img src="{{ url_for('video_feed') }}" width="100%">
```

Figure 5: Sample web page template in flask

V. RESULT

The result shows that recognize the face of a given user. Facial recognition has proven to be one of the most secure methods of all biometric systems to a point for high level security and to avoid ATM robberies and provide security for ATM. It will firstly authorize using the user face if it will match then it is moves to the user interface and send OTP and verified it.

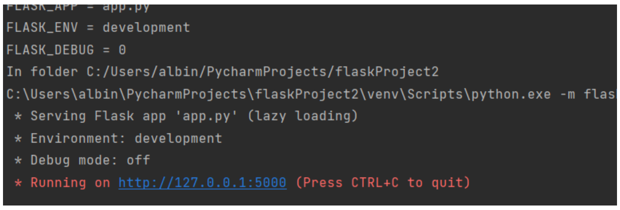

Figure 6: URL of web page

VI. CONCLUSION

This paper aims to use biometrics to make the ATM transaction structure increasingly reliable and secure. The idea of OTP and facial recognition added to the framework further improves the security. We are thus developing an ATM model that is more authentic by ensuring security using facial recognition software. In this project, we tried to propose a solution to the problem worry about the problem of fraudulent transactions through the ATM by biometrics which can only be made possible when the account holder is physically available. Thus, it cancels the cases of illegal transactions at ATM points without the knowledge from the genuine owner. The use of a biometric function for identification is strong and it is even more protected when another is used at authentication level.

VII. REFERENCES

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